Hrushikesh's Master's Thesis:

Autoparametric Resonance Based Energy Harvesting from Vortex Induced Vibrations

Primary Objective:

- Model a system that harvests energy from vortex-induced vibrations (VIV)
- Use autoparametric resonance to increase bandwidth and energy harvesting capability beyond lock-in (resonant) region of primary system

Autoparametric resonance:

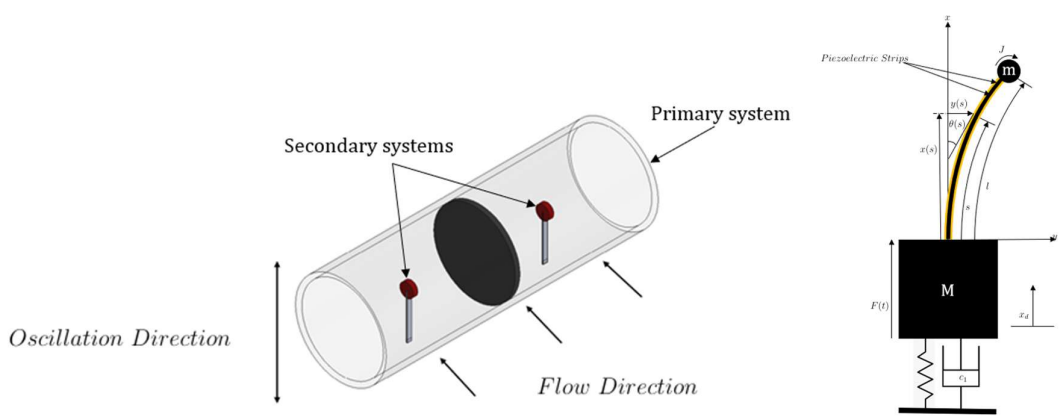
A nonlinear phenomenon relating to internal resonance where energy is transferred between coupled modes of a system, causing a secondary oscillator to become excited when the primary system is driven at a specific frequency ratio (typically twice the natural frequency of the primary system). The energy transfer in autoparametric resonance happens as a consequence of the motion of the system itself.

Methodology:

- Developed a nonlinear electromechanical model of a VIV-driven primary system coupled with frequency-tuned cantilever type secondary systems exhibiting autoparametric resonance
- Incorporated wake oscillator-based fluid forcing, piezoelectric coupling for energy harvesting and modelled a geometrically non linear system.
- Validated experimentally (vibration shaker and wind tunnel tests), demonstrating vibration suppression and broadband energy harvesting beyond the lock-in region

System Architecture:

- Primary system: Bluff body (cylinder) undergoing VIV
- Secondary system(s): Cantilever beams with end masses inside the cylinder
- multiple secondary oscillators for multiple autoparametric resonant frequencies



Representative Schematics

Mathematical Formulation:

A non linearly coupled model describes the dynamics of the motion of the primary and secondary systems, excited by fluid flow described using the semi empirical wake oscillator model

1) Primary system

$$\overbrace{\ddot{x} + 2\epsilon_{21}\mu_1\dot{x} + \omega_1^2 x}^{\text{Motion of the primary system}} \overbrace{- \epsilon_{21}\left(q_1\ddot{q}_1 + \dot{q}_1{}^2\right) - \epsilon_{22}\left(q_2\ddot{q}_2 + \dot{q}_2{}^2\right)}^{\text{Non linear Coupling terms}} =$$

$$\underbrace{\frac{1}{2(M + m_1 + m_2 + \bar{m}_1 + \bar{m}_2)}\rho_f U_\infty^2 DL\left(\frac{QC_l^0}{\hat{Q}} - \frac{C_d\dot{x}}{U_\infty}\right)\sqrt{1 + \left(\frac{\dot{x}_d X_0}{U_\infty}\right)^2}}_{\text{Forcing due to the free stream (Semi empirical wake oscillator model)}}.$$

2) Secondary System

$$\left.\begin{aligned} &\ddot{q}_1 + 2\epsilon_{21}\mu_{21}\dot{q}_1 + \omega_{beam1}^2 q_1 + \delta_{21}\epsilon_{21}^2 q_1^3 + R_1\epsilon_{21}^2 q_1\left(q_1\ddot{q}_1 + \dot{q}_1{}^2\right) - \epsilon_{21}\ddot{x}q_1 = 0. \\ &\ddot{q}_2 + 2\epsilon_{22}\mu_{22}\dot{q}_2 + \omega_{beam2}^2 q_2 + \delta_{22}\epsilon_{22}^2 q_2^3 + R_2\epsilon_{22}^2 q_2\left(q_2\ddot{q}_2 + \dot{q}_2{}^2\right) - \epsilon_{22}\ddot{x}q_2 = 0. \end{aligned}\right\} \text{Equations of motion for the secondary system}$$

$$\left.\begin{aligned} &\frac{V_{n1}(t)}{R} + C_p\dot{V}_{n1}(t) - \theta_{p1}\dot{q}_1(t) = 0. \\ &\frac{V_{n2}(t)}{R} + C_p\dot{V}_{n2}(t) - \theta_{p2}\dot{q}_2(t) = 0. \end{aligned}\right\} \text{Piezoelectric voltage equations}$$

3) Wake Variable (for the flow physics)

$$\ddot{Q} + \epsilon_x\omega_f((QX_0)^2 - 1)\dot{Q} + \omega_f^2 Q = \frac{A_x}{D}\ddot{x}$$

Note: Equations are non-dimensional.

For a detailed derivation and explanation on the parameters follow the link

Solution Methodology:

1) Used Mathematica for simplifying the derived symbolic equations and modal discretization
2) Calculated the natural frequencies and mode shapes of the beam using MATLAB
3) Calculated governing parameters of the equation based on the independent quantities
4) Decomposed equations into State Space formulation
5) Iteratively solved for the displacements and voltage using ode45 and ode15s solvers in MATLAB

Link to Code

Link to Thesis